

Microsoft Intune Endpoint Management Lab Report

Author: Shawn Roth **Date:** May 13, 2026 **Purpose:** Hands-on lab to build practical experience with Microsoft Intune for the Desktop Support II role at F5

Executive Summary

This lab demonstrates end-to-end Microsoft Intune configuration and security management. A Windows 11 Pro virtual machine was provisioned, enrolled, and managed using Intune policies, including custom firewall rules to block high-risk ports (445 & 80).

Key Outcomes:

- Successful device enrollment into Intune
 - Branded desktop configuration via Settings Catalog
 - Hardened security through Defender Firewall policies
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1. Lab Environment

- **Hypervisor:** VMware Workstation
 - **Operating System:** Windows 11 Pro (Virtual Machine)
 - **Tenant:** Microsoft 365 (labroth.onmicrosoft.com)
 - **Intune License:** Assigned to testuser@labroth.onmicrosoft.com
 - **Device Name:** INTUNELAB
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2. Lab Objectives

- Provision and enroll a Windows 11 endpoint
 - Create and apply Configuration Profiles
 - Implement custom Windows Defender Firewall rules
 - Verify policy application and device compliance
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3. Step-by-Step Implementation

3.1 VM Creation & Intune Enrollment

- Created persistent Windows 11 Pro VM
- Created test user with Intune license
- Enrolled device using **Access work or school** → MDM enrollment
- Verified device status in **Intune** → **Devices** → **All devices**

3.2 Configuration Profile (Lab Config)

- **Profile Type:** Settings Catalog
- **Settings Applied:**
 - Custom desktop wallpaper (labroth.co)
 - Prevent users from changing desktop background

3.3 Firewall Security Policies

Policy 1: Firewall Baseline

- Enabled Windows Defender Firewall for Domain, Private, and Public networks
- Default inbound action set to **Block**

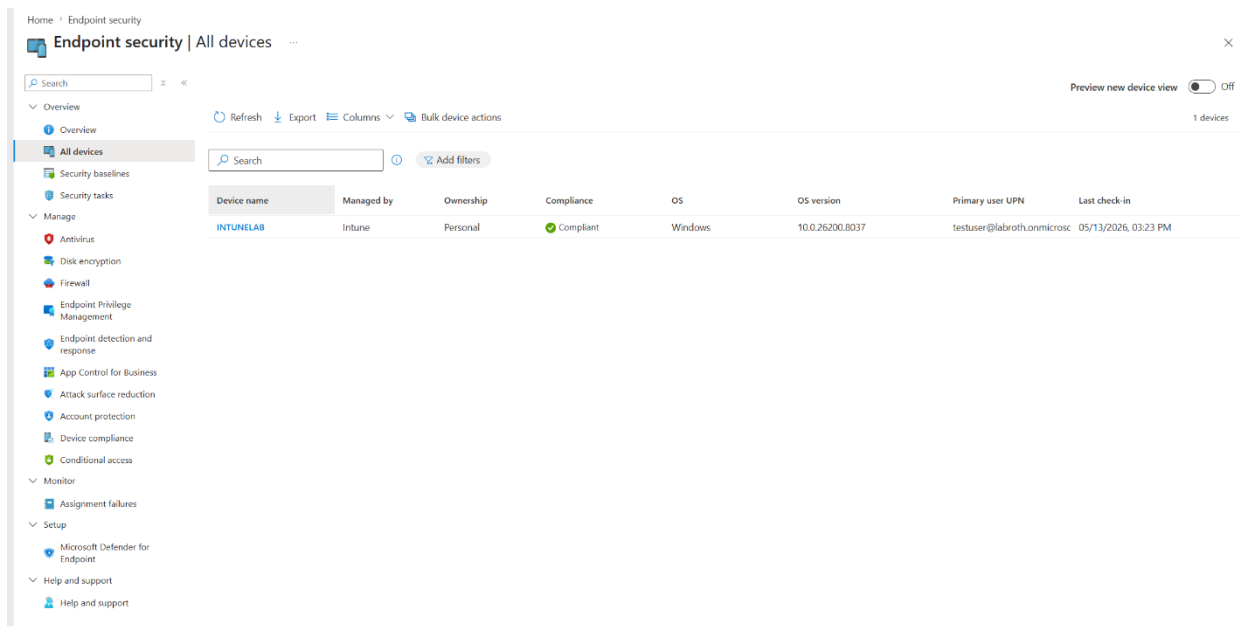
Policy 2: Custom Port Blocking Rules

- **Rule 1:** Block Inbound TCP Port **445** (SMB)
- **Rule 2:** Block Inbound TCP Port **80** (HTTP)
- Applied to all network profiles (Domain, Private, Public)

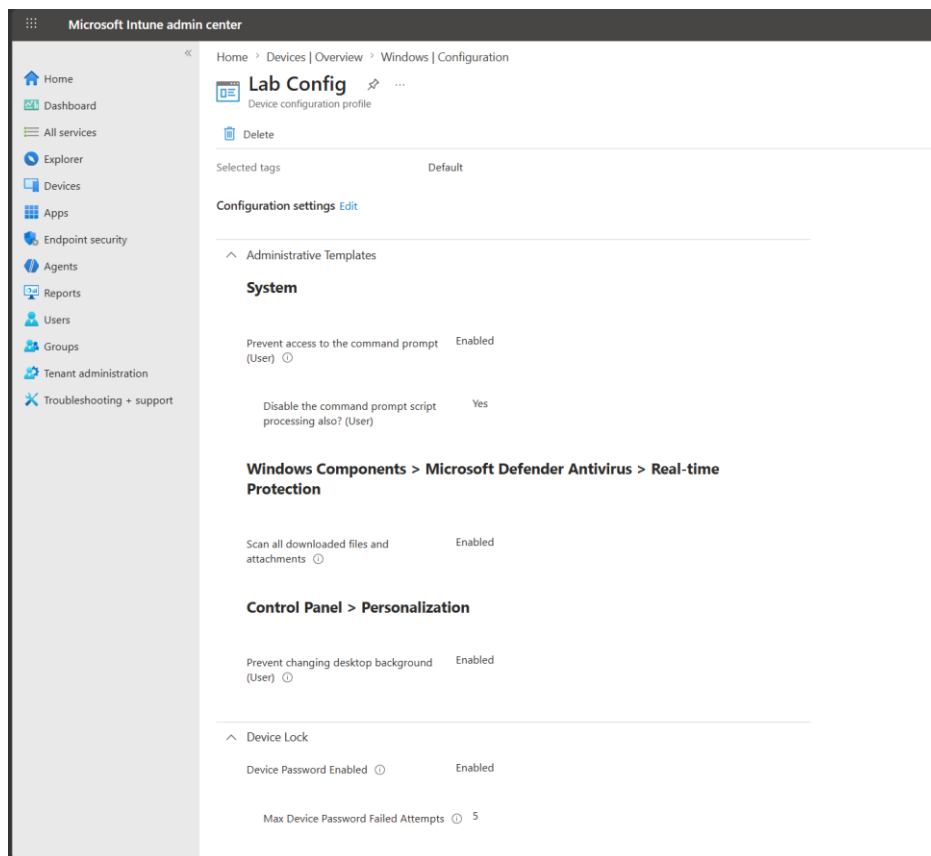
4. Screenshots

(Insert screenshots here — recommended order)

1. Enrolled device in Intune (All Devices)



2. Configuration Policies overview



3. Firewall Baseline Policy settings

Microsoft Intune admin center

Home

Dashboard

All services

Explorer

Devices

Apps

Endpoint security

Agents

Reports

Users

Groups

Tenant administration

Troubleshooting + support

Home > Windows | Configuration

Port config

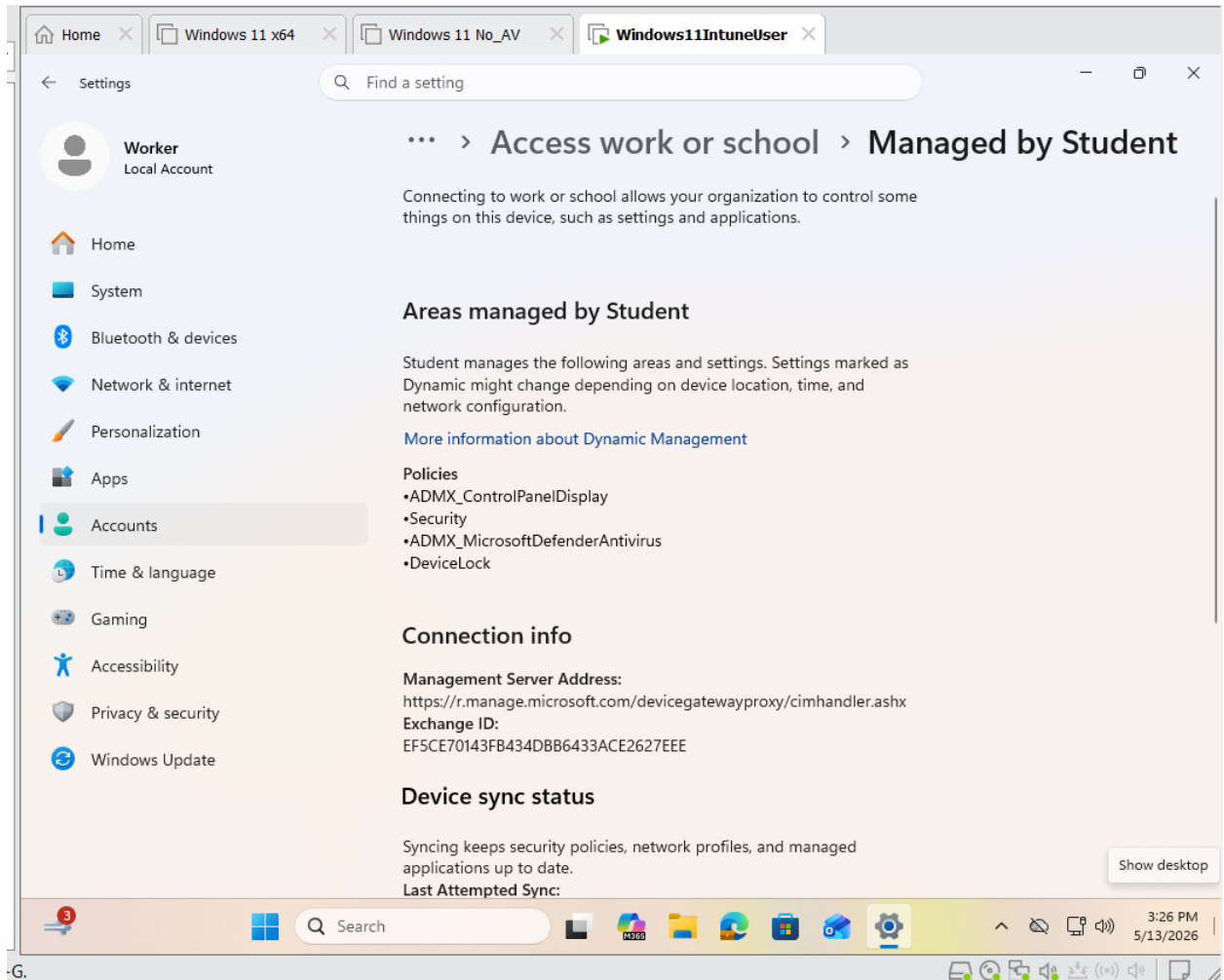
Device configuration profile

Delete

Action ⓘ	Block
Direction ⓘ	The rule applies to outbound traffic.
Enabled ⓘ	Enabled
Name ⓘ	Block outbound traffic on port 80
Interface Types ⓘ	All
Remote Port Ranges ⓘ	80

Enabled ⓘ	Enabled
Name ⓘ	Block inbound traffic on port 445 (SMB)
Interface Types ⓘ	All
Remote Port Ranges ⓘ	445
Local Port Ranges ⓘ	445
Action ⓘ	Block
Direction ⓘ	The rule applies to inbound traffic.

4. Custom Firewall Rules (showing blocked ports 445 & 80)



5. Device configuration status (showing applied policies)

The screenshot displays the Microsoft Intune admin center interface. On the left is a navigation pane with the following items: Home, Dashboard, All services, Explorer, Devices, Apps, Endpoint security, Agents (highlighted), Reports, Users, Groups, Tenant administration, and Troubleshooting + support. The main content area is titled 'Home > Windows | Configuration' and shows the 'Firewall config' page, which is a 'Device configuration profile'. At the top of the main area, there is a 'Delete' button and a 'Selected tags' section with a 'Default' tag. Below this is a 'Configuration settings' section with an 'Edit' link. The 'Firewall' section is expanded, showing a description: 'The Firewall configuration service provider configures the Windows Defender Firewall global settings, per profile settings, as well as the desired set of custom rules to be enforced on the device. Using the Firewall CSP the IT admin can now manage non-domain devices, and reduce the risk of network security threats across all systems connecting to the corporate network.' Below the description is a list of settings:

Enable Domain Network Firewall ⓘ	True
Enable Log Success Connections ⓘ	Disable Logging Of Successful Connections
Disable Stealth Mode ⓘ	False
Allow Local Ipsec Policy Merge ⓘ	True
Log File Path ⓘ	%systemroot%\system32\LogFiles\Firewall\pfirewall.log
Default Outbound Action ⓘ	Allow
Shielded ⓘ	False
Disable Inbound Notifications ⓘ	False
Auth Apps Allow User Pref Merge ⓘ	True

6. Final VM desktop with applied wallpaper



5. Verification & Testing

- Policies successfully assigned to device
- Verified firewall rules are active
- Device shows as **Compliant**
- Policy sync and application tested multiple times

6. Key Learnings

- Intune device enrollment process and troubleshooting
- Difference between Settings Catalog and Endpoint Protection profiles
- Creating and managing custom Windows Firewall rules in Intune
- Policy assignment, targeting, and sync behavior
- Importance of device activation state for certain policies

- Real-world security hardening techniques
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7. Conclusion & Future Work

This lab provided valuable hands-on experience with Microsoft Intune, directly relevant to modern Desktop Support and endpoint management roles.

Future Enhancements:

- Application deployment (Win32)
- BitLocker encryption policies
- Compliance policies with auto-remediation
- Conditional Access implementation